

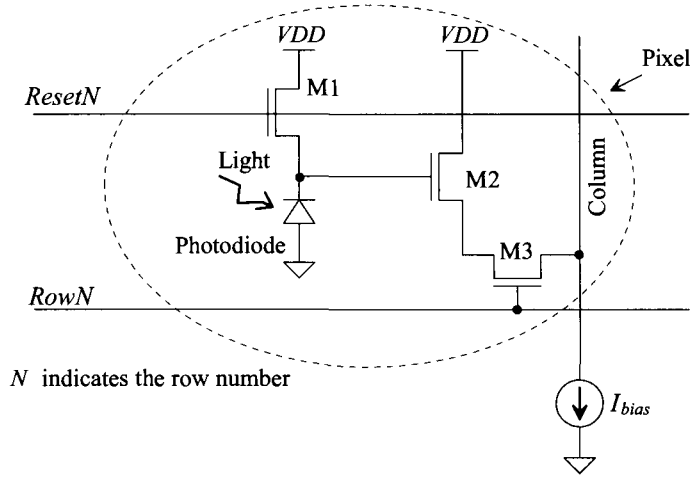


Figure 17.23 A CMOS active pixel sensor (APS).

the *ResetN* going low and the data (the column voltage, V_I , corresponding to the intensity of the light striking the photodiode) being sampled on the column line is called the *aperture time*. Note that a dark signal corresponds to a large voltage on the column line ($VDD - V_{THN}$), while a bright signal corresponds to a lower voltage (less than $VDD - V_{THN}$).

Sampling the Reference and Intensity Signals

When *ResetN* and *RowN* are high, the pixel's reference (or dark) voltage, V_R , is placed on the column line, Fig. 17.24. At this time the sample and hold reference signal, *SHR*, goes high and V_R is sampled onto a hold capacitor. Next, *ResetN* goes low and the photodiode changes light into charge. After the aperture time, the information from the photodiode (the intensity of the light) is on the column line, V_I . This voltage is then sampled and held on a hold capacitor when the signal *SHI* (sample and hold the intensity of the light) goes

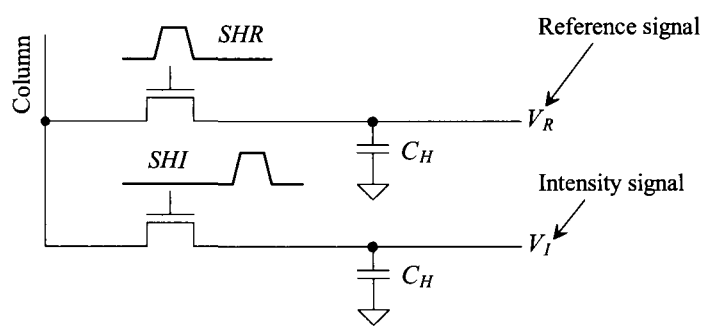


Figure 17.24 Sampling the reference and intensity signals.

high. What we want is to design a circuit that takes the difference in V_R and V_I and generates a digital number.

Noise Issues

Circuit noise limits the dynamic range of the sense, resulting in the blurring of the images or the inability to detect low light levels or distinguish between bright or high-intensity light levels. The major noise sources in CMOS, as discussed in Chs. 8 and 9, are flicker and thermal noises. The first design value we must select is the size of the hold capacitor in Fig. 17.24. The size of this capacitor limits the thermal noise in the sample. As seen in Table 8.1, using a hold capacitor of 1 pF results in an RMS noise voltage, just due to thermal noise, of 64 μ V. This corresponds to a peak-to-peak voltage in the time domain of roughly 400 μ V (six times the RMS value as seen in Fig. 8.33). Using a larger capacitor takes up more layout area and takes longer to charge but lowers the amount of thermal noise sampled onto C_H .

The output current of the pixel (when M3 is on and M2 is a source follower) also contains flicker noise. When this pixel is connected to the large hold capacitance and the capacitance of the bit line, the flicker noise current will be integrated. The result is a noise power spectral density with a $1/f^3$ spectral shape. As discussed in Ex. 8.14, the RMS value of the resulting noise signal will grow linearly with measurement time. What this means is that to *achieve a low noise* sample onto C_H we want to minimize the amount of time *SHR* and *SHI* are high *and* the time difference between the two signals. *SHR* and *SHI* should only go high long enough to charge C_H . The amount of time between M1 turning on and *SHI* shutting off should be as small as possible.

The amount of noise in V_R and V_I cannot be reduced after they are captured on the hold capacitors. For example, ideally V_I may be 0.5 V. However, because of noise V_I may be 0.501 V or 0.495 V, etc. We *may* be able to reduce the noise by averaging successful samples though (see Eq. [8.48] and Ex. 8.14).

Ideally, the sensing circuitry (the circuitry used to change the analog voltages, V_R and V_I , into a digital number) doesn't introduce any additional noise. When using the DSM, the counter can be thought of as a lowpass digital filter, Fig. 17.25. (See also the book entitled *CMOS Mixed-Signal Circuit Design* for much more detailed description of the frequency response of a counter.) As seen in this figure, if the sense time is the total number of times the DSM is clocked, N , multiplied by the clock's period T , that is $T_{meas} = NT$, then increasing the sense times lowers the bandwidth of the digital filter. This has the effect of lowering the noise bandwidth (so that the DSM sensing circuit does not contribute any further noise to the measured signal).

An example where a counter is used that doesn't result in filtering the measured signal is seen in Fig. 17.26. Here the column voltage corresponding to the intensity of the light is used as the $-$ input to a comparator. For the $+$ input, a constant current is used to charge a capacitor to generate a voltage ramp. When the ramp's voltage gets above V_I , the output of the comparator goes high and stops the counter. In this scenario the counter is simply used to give a digital representation of a time or ramp voltage. The counter doesn't provide any filtering (or more precisely averaging as used in DSM, see Tables 17.1 to 17.3). Notice also that this sensing topology very likely adds noise to the measured signal. For example, we know that using a constant current to charge a

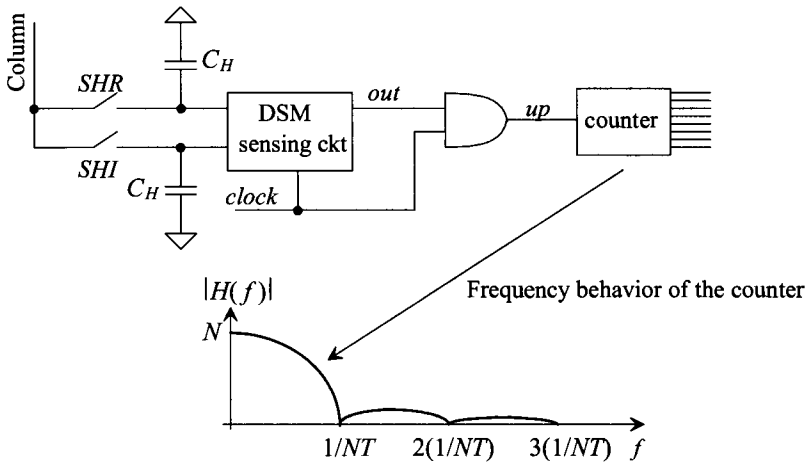


Figure 17.25 Thinking of the counter as a digital low pass filter

capacitor results in a voltage signal containing $1/f^3$ noise. The ramp, with noise, voltage is connected directly to the comparator and causes noise to be added to the measured signal. Further, if the comparator makes an error and switches states too early or too late because of an offset or noise coupling from the adjacent column sensing circuits, again, the sense adds noise to the measured signal (the digital output code isn't constant but rather moves around even though the inputs to the sense amplifier may be constant).

Also note that using the DSM, we can run the sense operation indefinitely, while the scheme in Fig. 17.26 is limited by the ramp rate. Further increasing the clock frequency in Fig. 17.26 won't increase the resolution if the sense is noise limited. In the DSM sensing circuit, increasing the clock frequency, as seen in Fig. 17.25, lowers the bandwidth of the digital filter and increases the resolution of the sense.

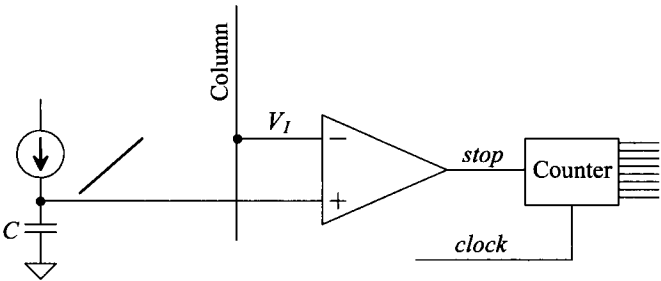


Figure 17.26 A circuit where the counter doesn't behave like a lowpass filter.

Subtracting V_R from V_I

As we mentioned earlier, each pixel has slightly different characteristics. For example, the threshold voltage of M2 in one pixel may be 10 mV different from the threshold voltage of M2 in a different pixel. To remove this error, we subtract the measured reference voltage, V_R , from the measured signal intensity, V_I . It's important, during the sense, not to change these voltages with our sensing circuit. Let's convert these voltages to currents and then subtract the currents to get the difference (and not try to subtract the voltages directly).

Examine the voltage-to-current converter seen in Fig. 17.27. This circuit is simply a source follower that is made wide so that its V_{SG} is always approximately the threshold voltage. The relationship between the drain current and the column voltage V_{col} (V_I or V_R) is

$$I = \frac{VDD - V_{THP} - V_{col}}{R} \quad (17.30)$$

for

$$V_{col} < VDD - V_{THP} \quad (17.31)$$

Reviewing Fig. 17.23, we see that if the threshold voltage drop of M2 (with body effect) is more than V_{THP} (without body effect because we placed the PMOS device in Fig. 17.27 in its own well), Eq. (17.31) will always be satisfied.

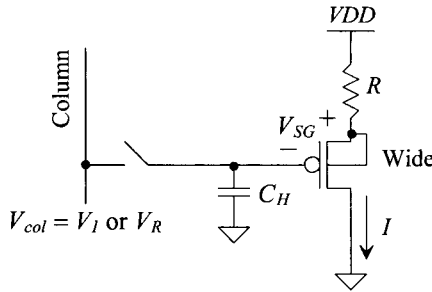


Figure 17.27 Linear voltage-to-current conversion.

Example 17.6

Using SPICE determine the linearity of the current in the circuit seen in Fig. 17.28. Use the short-channel CMOS process and compare the simulation results to hand calculations.

If VDD is 1 V, $V_{THP} = 250$ mV, and R is 10MEG, then

$$I = 75 - 100 \cdot V_{col} \text{ nA}$$

Noting that the ideal slope of the line is $-1/R (= -100 \times 10^{-9})$. Figure 17.29 shows the simulation results. In (a) we see from the transfer curve how the output, I , changes with the input, V_{col} . In (b) we take the derivative of I to see if the slope is perfectly flat. The linearity is pretty good (1%) for $V_I < 400$ mV. ■

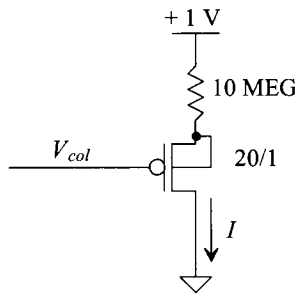


Figure 17.28 Circuit used in Ex. 17.6.

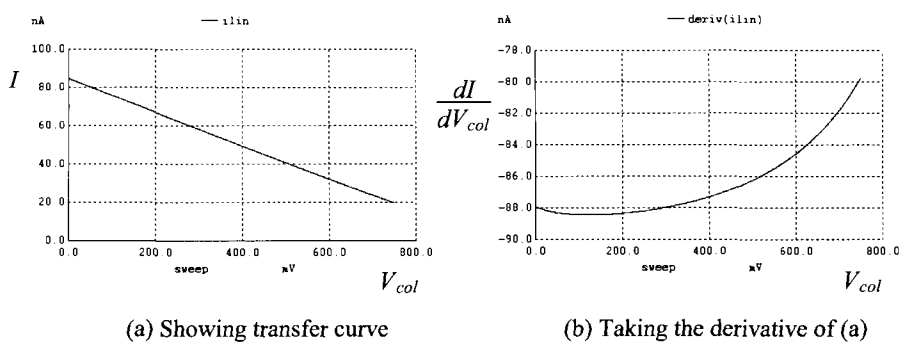


Figure 17.29 Simulating the operation and linearity of the voltage-to-current converter in Fig. 17.28.

We might wonder, from the last example, if it would be a good idea to try to make the linearity of our voltage-to-current converter even better. Before doing this, let's look at the linearity of the source follower in the pixel itself. Figure 17.30 shows a simplified schematic of the source follower (M2) used in the APS. We've modeled the finite output resistance of the current source with a 10MEG resistor. The simulation results in Fig. 17.31 show the transfer relation of this circuit and its linearity. For column voltages between 100 mV and 600 mV, the linearity is 0.5% (which is comparable so we won't concern ourselves any further with trying to better the linearity at this point).

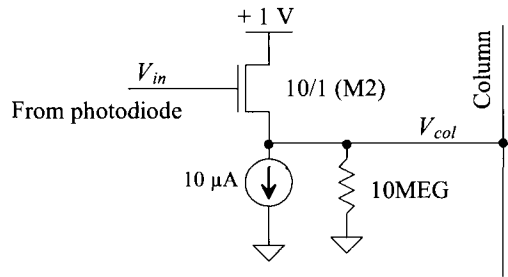


Figure 17.30 Looking at the linearity of the source follower in the pixel.

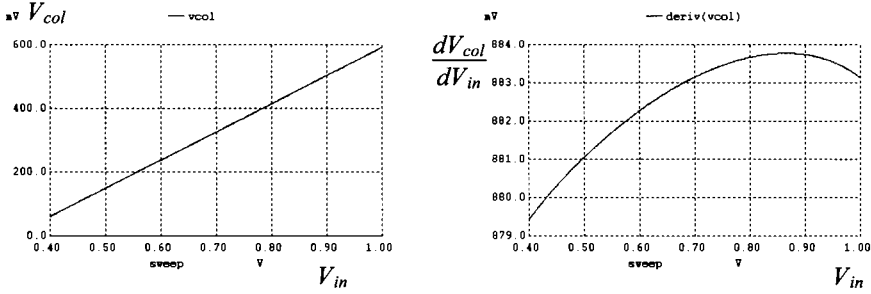


Figure 17.31 Showing the transfer curve and linearity of the circuit in Fig. 17.30.

To take the difference in the currents, let's use a current mirror as seen in Fig. 17.32. The current corresponding to the reference voltage is

$$I_R = \frac{V_{DD} - V_{THP} - V_R}{R_R} = \frac{V_{R,shift}}{R_R} \quad (17.32)$$

and the current corresponding to the intensity of light is

$$I_I = \frac{V_{DD} - V_{THP} - V_I}{R_I} = \frac{V_{I,shift}}{R_I} \quad (17.33)$$

The difference in these currents is summed (sigma) in the bucket capacitor, as seen in Fig. 17.32. When we add our comparator and feedback to form a DSM, the charge on the capacitor, averaged over time, is a constant. This occurs when

$$I_R = I_I = \frac{V_{R,shift}}{R_R} = \frac{V_{I,shift}}{R_I} \quad (17.34)$$

or

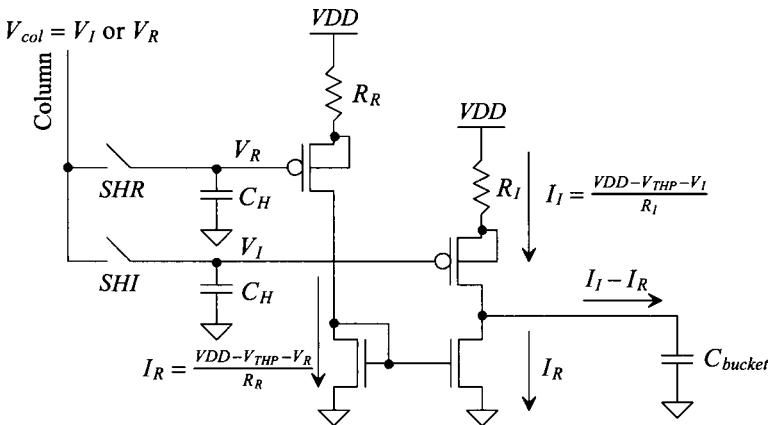


Figure 17.32 Subtracting the currents.

$$V_{I,shift} = \frac{R_I}{R_R} \cdot V_{R,shift} \quad (17.35)$$

The ratio of the resistances gives us the information we need to determine the relative (to the reference level, V_R) intensity of light on the pixel. To implement the resistors, let's use the switched-capacitor resistor as seen in Fig. 17.20. For the reference voltage we know

$$V_{I,shift} \geq V_{R,shift} \quad (17.36)$$

and so ($R_I \geq R_R$). Let's use (M3 always on in Fig. 17.20)

$$R_R = \frac{1}{f \cdot C_{cup}} \quad (17.37)$$

and for R_I (which will be enabled via M3 when its gate goes low)

$$R_I = \frac{1}{f \cdot C_{cup} \cdot \frac{\bar{M}}{N}} \quad (17.38)$$

Rewriting Eq. (17.35), knowing $\bar{M} = N - M$, gives

$$V_{I,shift} = V_{R,shift} \cdot \frac{N}{N - M} \quad (17.39)$$

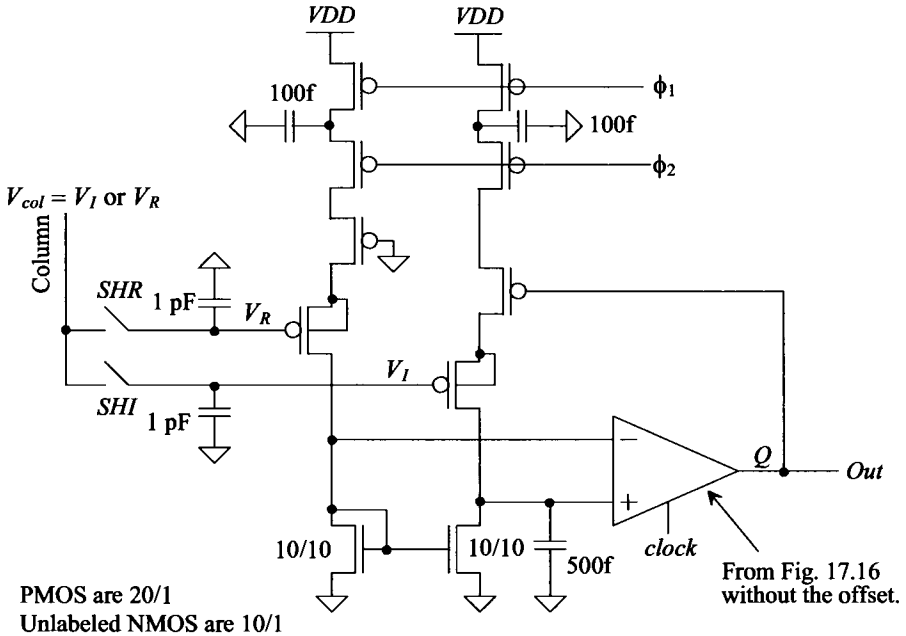
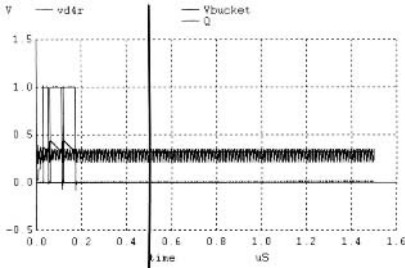


Figure 17.33 Schematic of a DSM for sensing in CMOS imaging chips.

If the DSM is clocked 1,000 times (N), then M can range from 0 (the pixel is not illuminated, that is, the dark or reference level) to 1,000 (very bright). Figure 17.33 shows a schematic of the sensing circuit with some typical values.

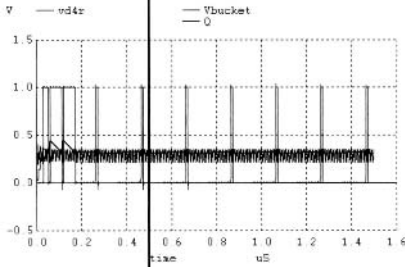
The circuit in Fig. 17.33 was made as symmetrical as possible so that power supply or ground noise affected each signal path equally. The MOSFETs in the current mirror are made long (10 drawn) so that the voltages on the input of the comparator are greater than the NMOS threshold voltage. The size of the capacitors isn't that critical. We want the C_{cup} capacitors to be less than the C_{bucket} capacitor. We don't have to worry about overcharging C_{bucket} in this scheme because the input signal contributions are limited by the switched capacitor resistors. (We only get one C_{cup} every clock cycle.)

Figure 17.34 shows the simulation results for the DSM circuit of Fig. 17.33. We assume that the counter, Fig. 17.25, is enabled after 500 ns. Prior to this time, the



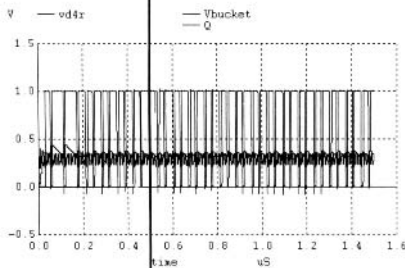
(a)

$$\begin{aligned} V_R &= 650 \text{ mV} & V_I &= 650 \text{ mV} \\ V_{R,shift} &= VDD - V_{THP} - V_R = 100 \text{ mV} \\ V_{I,shift} &= VDD - V_{THP} - V_I = 100 \text{ mV} \\ V_{I,shift} &= 100 \text{ mV} \cdot \frac{100}{100-0} = 100 \text{ mV} \quad (\text{sensed}) \end{aligned}$$



(b)

$$\begin{aligned} V_R &= 650 \text{ mV} & V_I &= 645 \text{ mV} \\ V_{R,shift} &= VDD - V_{THP} - V_R = 100 \text{ mV} \\ V_{I,shift} &= VDD - V_{THP} - V_I = 105 \text{ mV} \\ V_{I,shift} &= 100 \text{ mV} \cdot \frac{100}{100-5} = 105 \text{ mV} \quad (\text{sensed}) \end{aligned}$$



(c)

start sensing at 500 ns

$$\begin{aligned} V_R &= 650 \text{ mV} & V_I &= 400 \text{ mV} \\ V_{R,shift} &= VDD - V_{THP} - V_R = 100 \text{ mV} \\ V_{I,shift} &= VDD - V_{THP} - V_I = 350 \text{ mV} \\ V_{I,shift} &= 100 \text{ mV} \cdot \frac{100}{100-72} = 357 \text{ mV} \quad (\text{sensed}) \end{aligned}$$

Figure 17.34 How the DSM sensing circuit in Fig. 17.33 operates.

reference and intensity signals are sampled onto the hold capacitors. If $V_R = 650$ mV, then

$$V_{R,shift} \approx 1 - 0.25 - 0.65 = 100 \text{ mV}$$

In Fig. 17.34a we apply the same signal to the DSM sensing circuit, that is 650 mV, for V_I . As expected, the output stays low for all times. In (b) we drop V_I to 645 mV and see, during the 500 ns to 1,500 ns sensing time, 5 output ones. As seen in the figure, the sensed value indicates the intensity is 5 mV below the reference. In (c) we drop the intensity signal to 400 mV resulting in $V_{I,shift} = 350$ mV. The sensed value with 72 of the 100 clock cycles being high is 357 mV. Let's look at what would happen if we sensed 71 ones

$$100 \text{ mV} \cdot \frac{100}{100 - 71} = 345 \text{ mV}$$

In either case (71 or 72 ones) the resolution was so coarse that we couldn't resolve the actual signal. If we think about this for a moment, we see that if the counter output code is small then the resolution of the measurement is better. For example, counter outputs of 1 and 2 result in

$$100 \text{ mV} \cdot \frac{100}{100 - 1} = 101 \text{ mV} \text{ and } 100 \text{ mV} \cdot \frac{100}{100 - 2} = 102 \text{ mV}$$

Looking at Eq. (17.39), we see that the dependence on M (the number of times the output of the DSM goes high) is *not linearly related* to the light intensity, V_I . What we want is an equation like (17.7), that is,

$$V_{I,shift} = V_{R,shift} \cdot \frac{M}{N} \quad (17.40)$$

To get a relationship like this, we might try to control the value of R_R too as seen in Fig. 17.35. The complementary output of the comparator is fed back so that

$$R_R = \frac{1}{f \cdot C \cdot \frac{M}{N}} \quad (17.41)$$

and thus

$$V_{I,shift} = V_{R,shift} \cdot \frac{M}{N - M} \quad (17.42)$$

Again, we do not have a linear relationship. Further, half or more of our outputs must be used to enable the switched-capacitor resistor in series with the reference signal. If, for example, $V_{R,shift} = V_{I,shift}$, then $M = N/2$. Since $V_R \geq V_I$, then $0 \leq M \leq N/2$ (not good design).

If we review the derivations leading up to Eqs. (17.7) and (17.20), the common theme is that the feedback signal controlled by the comparator is a constant addition to the capacitive bucket (or bit line). In Fig. 17.18, for example, we used M4 to ensure that the charge from C_{cup} was a constant added to C_{bit} . In our current sense amplifier, Fig. 17.33, the signal we feedback is not a constant but rather a function of V_I . When sensing in a CMOS imager, the signal fed back should be a function of the reference level, V_R . The comparator's output should control a fed-back signal that is derived from V_R .

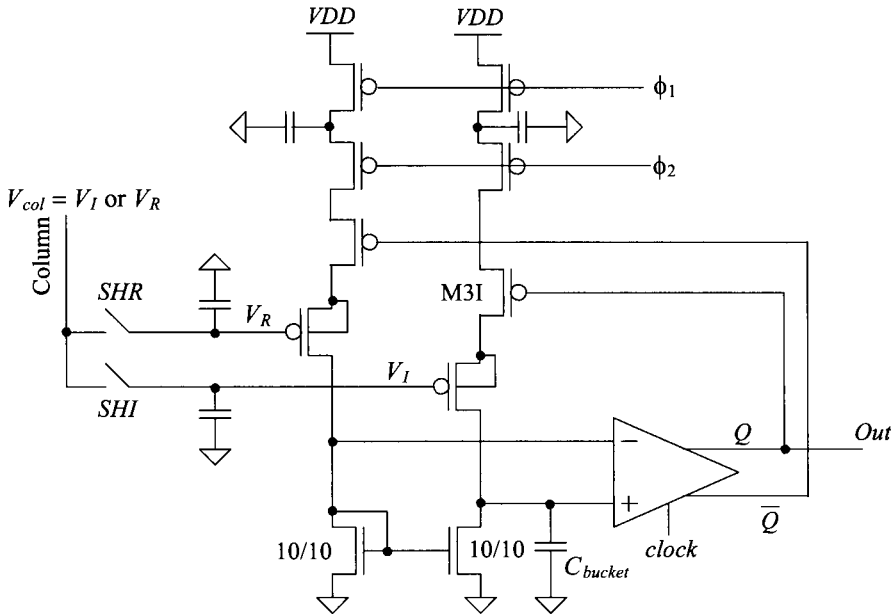


Figure 17.35 Using both comparator outputs for a DSM sensing circuit.

Note that it's a bad idea to ground the gate of M3I in Fig. 17.35 and only have a single feedback path. Since $V_R > V_I$, we won't be able to supply enough current through the reference signal path. The charge supplied by the intensity path will always be greater than the charge supplied from the reference path (and so C_{bucket} will overflow). A new topology is needed.

Figure 17.36 shows an NMOS version of Fig. 17.33. We've replaced the capacitors with MOSFETs to show that the DSM sensor can be implemented using a single-poly CMOS process. The PMOS devices are used for the "cup" capacitors. We use PMOS instead of NMOS because, for the topology seen, the PMOS devices always remain in strong inversion. The NMOS devices, for example, move towards accumulation mode when ϕ_2 turns on and discharges the capacitors. The result is a nonlinear capacitance (the size of the cup varies). Similarly, we use NMOS for the bucket capacitor because, for the topology used, they will always remain in strong inversion (the capacitance won't vary with the changes in the voltage on their gates). Note that a second "bucket" capacitor was added across the 10/10 diode connected (gate and drain tied together) PMOS device. This addition serves two purposes. The first is to ensure that ground noise affects the comparator inputs equally (noise on ground will feed evenly through the bucket capacitors to the input of the comparator). The second reason is that it smoothes out the summation of the currents. Note that if the added capacitor is too big, stability can be an issue (the added capacitor adds a delay in series with the feedback path).

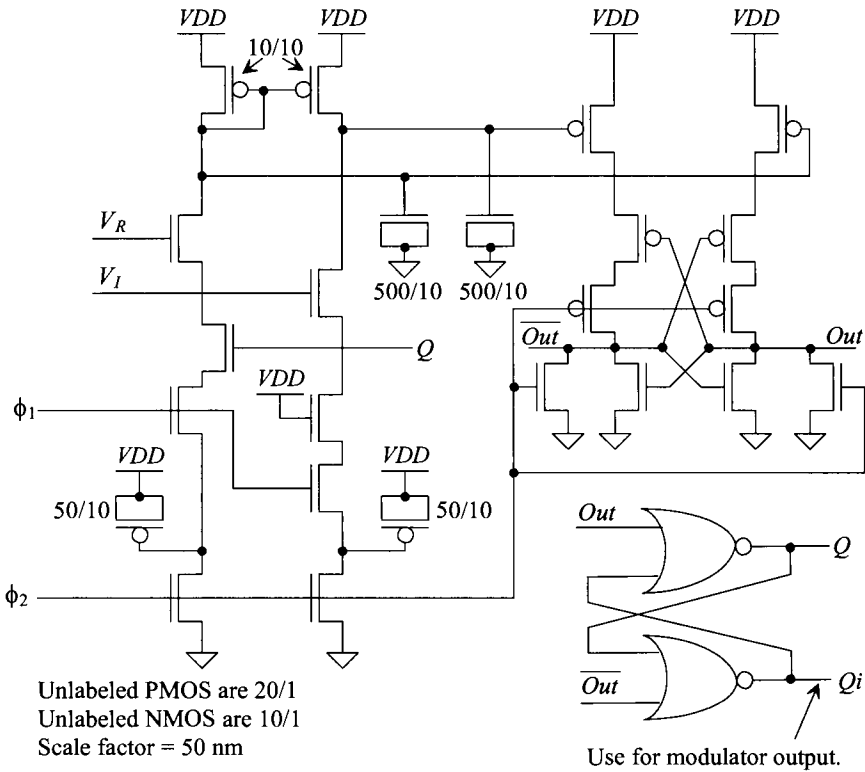


Figure 17.36 DSM circuit for sensing in a CMOS imaging chip.

In Fig. 17.36 we show a comparator design using PMOS imbalance MOSFETs (as seen in Fig. 16.32 for the NMOS flavor). When ϕ_2 goes high, the outputs of the comparator, *Out* and *Out*, are driven low. When ϕ_2 goes low, on the falling edge, the comparator makes a decision concerning which of the bucket capacitors has the higher potential across it. Based on this decision, the outputs of the comparator are latched with the NOR-based SR latch. The *Q* output is fed back to enable or disable the summation of charge via the reference path. We've used the *Qi* output as the DSM's output. When the intensity level is the same as the reference level, the output stays low for all times. It may be possible to eliminate this SR latch in a production part. However, we include it here because it makes the simulation results easier to look at (the glitches in the comparator's output are reduced).

Figure 17.37 shows some simulation results for the DSM circuit seen in Fig. 17.36. Notice, in this figure, that we've *assumed* the threshold voltage of the NMOS device is 250 mV. Since our reference level (the black level for the pixel) is 650 mV, our shifted reference level, $V_{R,shift}$, is 400 mV. For 100 samples then we can estimate the resolution as

$$V_{res} = \frac{V_{R,shift}}{N} \rightarrow 4 \text{ mV} \quad (17.43)$$